

STREAMING MEMORY

1 Item arrives

extract item + type
create K probes
insert into store

2 Serve & meter

retrieve top-k
count uses m_i
forecast reuse $\hat{n}_i$

3 Option gate

price trial value
compare write cost
enqueue if positive

4 Batch write

consolidate queue
LoRA + replay
meter write FLOPs

5 Fresh verification

closed-book re-probe
measure realized α_i
keep or roll back

PASS: keep in weights

price the next option

ROLLBACK → external store

update empirical state

Deterministic gates

candidate: $m_i \geq k_i^*$
trial: $\hat{n}_i \tilde{r}_i \geq \lambda C_i + \hat{\rho} / \bar{b}$
keep: $\alpha_i \geq \kappa_i = h - \lambda c_i / g_i$

Measured policy state

$F(\alpha)$	$\hat{h}, \hat{\rho}$	$\hat{n}_i$
verified write efficacy <i>trial outcomes</i>	retrieval + interference <i>anchor probes</i>	semantic reuse forecast <i>query history</i>